

Chapter 4

Coordinated Distributed Experiments (CDEs) Applied to Earthquake Forecast Test Sites



Zhongliang Wu, Yan Zhang and Jiawei Li

Abstract Field experiments for testing the hypotheses related to earthquake preparation and earthquake forecast have been facing dual challenges due to the nature of the study of earthquakes. On the one hand, physical laws derived from laboratory experiments, when applied to field, have the problems of scaling, which necessitate field experiments. On the other hand, due to the limitation of earthquake ‘samples’ and the complicated factors controlling the preparation of an earthquake, the tests of the physical hypotheses by field experiments have vague significance. Such challenges are not only in the field of earthquake studies. In ecology and environmental science, ‘few samples, many factors’ is also one of the difficulties blocking the test of scientific hypotheses. To tackle this problem, in those fields, ‘Coordinated Distributed Experiments (CDEs)’ was proposed as an operational tool for hypothesis testing. Such an idea also provides earthquake studies with a new vision. In connection to the top-level design of the China Seismic Experiment Site (CSES), in this chapter, we discuss the concept CDEs applied to the test sites of earthquake forecast. We propose that an ‘earthquake rupture scenario’ be used for the coordination.

Keywords Natural laboratory · Earthquake science · Coordinated Distributed Experiments (CDEs)

Z. Wu (✉)

Institute of Earthquake Forecasting, China Earthquake Administration, Beijing 100036, China
e-mail: wuzl@cea-igp.ac.cn

Y. Zhang · J. Li

Institute of Geophysics, China Earthquake Administration, Beijing 100081, China

J. Li

School of Earth and Space Sciences, Peking University, Beijing 100871, China

© Higher Education Press and Springer Nature Singapore Pte Ltd. 2019

Y.-G. Li (ed.), *Earthquake and Disaster Risk: Decade Retrospective of the Wenchuan Earthquake*, https://doi.org/10.1007/978-981-13-8015-0_4

4.1 Natural Laboratories in Earthquake Science

Field experiments have been proposed for decades in earthquake science. Related to earthquake forecast, there were several field experiments conducted, such as the earthquake prediction experiments in the former Soviet Union (Semenov 1969), the Parkfield experiment in the USA (Bakun and Lindh 1985; Roeloffs and Langbein 1994; Roeloffs 2000), the TDP experiment in Turkey (Evans et al. 1987), the Tokai experiment in Japan (Davis and Somerville 1982; Mogi 2004),¹ and the PRENLAB² project in the Iceland, among others. On the other hand, however, in the view of physics, such ‘experiments’ are to much extent problematic due to the limitation of the precision of the measurement and the lack of controllable processes.

Such situation is facing a historic change. In recent years, innovative technologies have been reshaping the horizon of seismological, geodetic, and geological studies. One of the examples is the temporal variation of Earth structure. As early as in the 1960s, pre-earthquake variation of velocity of seismic waves was claimed and discussed (e.g., Semenov 1969), providing earthquake forecast with optimistic clues. However, due to the limitation on the precision of the field measurement, the scientific conclusions about the precursory structural variation have not been on a solid ground. Since the twenty-first century, emerging technologies such as waveform interferometry have enhanced significantly the capability of observing the co-seismic and post-seismic variations of structures, which potentially makes it feasible to constrain the pre-earthquake structural variation (e.g., Wegler et al. 2009; Chao and Peng 2009; Audet 2010). Such new opportunities are noticeable in the study of earthquake preparation and in turn reactivate the interests in the field experiments.

Accordingly, on 12 May, 2018, at the opening ceremony of the International Conference for the Decade Memory of the Wenchuan earthquake with the 4th International Conference on Continental Earthquakes (4th ICCE), it was officially announced that the China Seismic Experiment Site (CSES) be established in the Sichuan-Yunnan region. This natural laboratory, as a continuation and expansion of the West-Yunnan Earthquake Prediction Experiment Site since 1980 and the National Experiment Site for Earthquake Monitoring and Forecast since 2014, aims to make use of the cutting edge science and technology to deepen the understandings of the preparation and occurrence of continental earthquakes and the cause of seismic disasters. The discussion in this chapter is in connection with the top-level design of the CSES.

¹Hoshiba (2006).

²Stefansson et al. (1998, 2001).

4.2 Challenges of the Natural Laboratories

Associated with the natural laboratories for earthquake science, for example, the being built CSES, what will be the most important challenge? Using the cliché of some international organizations ‘opportunities and challenges’: the main challenge is apparently that the ‘opportunity’ to ‘meet’ the ‘target earthquake’ is small. Indeed in the history of earthquake forecast experiments, for most of the test sites, the ‘target earthquake’ occurred either out of the range of the test area (such as the 1996 Lijiang earthquake which is near to, but just out of, the West-Yunnan Earthquake Prediction Experiment Site), or within the test area but after the test has formally closed (such as the 2004 Parkfield earthquake which occurred long after the predicted earthquake, 1988 ± 4 years, and the planned duration of the experiment, 1984–1992, although observational facilities were still functioning). Ironically, the failure of the earthquake forecast test sites, which were originally designed for the test of short-term earthquake forecasts, was in fact due to the uncertainties of intermediate-term earthquake forecast.

Taking a larger area and longer duration of the test might be an operational counter-measure against the uncertainty of the intermediate-term earthquake forecast. However, the opportunity to ‘meet’ an earthquake is still problematic. In the decade summary of the Collaboratory for the Study of Earthquake Predictability (CSEP) project, it was pointed out that ‘*the dearth of large earthquakes in individual regions is a major limitation of evaluations*’ (Schorlemmer et al. 2018).

Another important issue is the complicated factors controlling the process of the preparation of an earthquake, which necessitates and challenges the field experiments. In the above sentence, ‘necessitate’ means that field experiments are important as compared to laboratory experiments—just recall the ‘heat flow paradox’ (e.g., Lachenbruch et al. 1995; Mora and Place 1998); and ‘challenge’ means that field experiments, even if with good design, always have to face to the fact that more factors seem to be determining the preparation process of an earthquake, and actually most if not all of the factors cannot be controlled.

4.3 Coordinated Distributed Experiments (CDEs)

It is worth pointing out that the difficulties discussed in the above section are not only in the field of earthquake studies. In ecology and environmental science, for example, ‘few samples, many factors’ is also one of the difficulties blocking the test of scientific hypotheses. Comparing to ecology and environmental science which have the difficulty in obtaining the laws at the global scale from individual local studies, earthquake science faces the difficulty in obtaining the regularities of earthquake preparation and occurrence from the studies with the duration much shorter than an earthquake cycle. To some extent, the ‘sampling’ problem is mainly ‘spatial’ for ecology and environmental science, and mainly ‘temporal’ for earthquake

science. On the other hand, however, it is common for ecology/environmental science and earthquake science that the ‘samples’ are not sufficient, and there are many controlling parameters to be considered in the test of the hypotheses.

To tackle this problem, in the fields of ecology and environment science, the concept ‘Coordinated Distributed Experiments (CDEs)’ was proposed as an operational tool for hypothesis testing (Fraser et al. 2012). Similarly, in earthquake science, we may consider the use of the spirit of the CDEs, although the forms would be different.

The CDEs in ecology have the following features (Fraser et al. 2012): (1) hypothesis-driven experimental study; (2) multiple geographic locations; (3) standardized research design; (4) standardized data and coordinated data management; (5) intellectual property sharing; (6) synchronized data collection; (7) multiple investigative teams; and (8) low cost and low maintenance. In the above (1)–(5) are called ‘defining attributes’, and (6)–(8) ‘probable characteristics’. It can be seen that almost none of the above-mentioned features are new to earthquake science. Earthquake science is even more advanced regarding ‘standardized data and coordinated data management’.

In the perspective of the CDEs for earthquake science, there is a need of not only one or two but also several test sites with different tectonic settings. This is not a new idea either. In 1997, the International Association of Seismology and Physics of the Earth’s Interior (IASPEI) passed the resolution³ related to the test areas for earthquake prediction:

- **Recognizing** that research into earthquake prediction needs to be carried out over long time scales with extensive and detailed observation at substantial cost, and **aware** that many nations face serious threats to their populations with limited resources and skills,
- IASPEI **urges** the organization of additional multinational test areas in different tectonic settings where high-level research efforts are already in progress, for example, Kamchatka (plate-subduction), Iceland (plate spreading), Yunnan (intercontinental strike-slip), Gulf of Corinth (continental rifting), Beijing (intra-continental)
- and **recommends** that host countries welcome participants from all nations and in due course make the data available to the international research community in computer-accessible form.

Earlier, in 1991, IASPEI passed the resolution which stressed almost the same idea:

- **Recognizing** the importance of earthquake source and prediction research to better understanding of the causes of devastating earthquakes and to work toward predicting them,
- IASPEI **encourages** national organizations supporting earthquake research to give high priority to funding of work in the IASPEI-approved sites for international collaborative earthquake prediction research, which are: (1) North Anatolian strike-slip fault (Turkey), (2) Aleutian-Alaskan subduction zone (the USA), (3) deep

³<http://iaspei.org/about/resolutions-statements>, last access: July 31, 2018.

mine-induced seismicity (South Africa), and (4) intermediate intraplate earthquakes (Vrancea, Romania) in the framework of the circum-Pannonian Carpathian seismic belts.

But the resolutions did not specify how to ‘approve’ and how to conduct the field test. To some extent, it had already had the concept ‘Distributed’, but not yet the concept ‘Coordinated’ at that time. It turns out that for the natural laboratories of earthquake science, ‘DEs’ without ‘C’⁴ might be one of the problems at the early time of the field experiment.

4.4 Guidelines for the Coordination

The Collaboratory for the Study of Earthquake Predictability (CSEP) was one of the significant advancements in implementing the ‘C’ for the field experiments. In the CSEP project, several testing areas were selected with standardized data format and testing criteria. It was pointed out that ‘*Meaningful evaluations of hypotheses about the long-term behavior of large earthquakes may take decades or centuries in regional fault systems, necessitating global models for testing hypotheses such as characteristic earthquakes, segmentation, and quasi-periodic recurrences*’ (Schorlemmer et al. 2018). The first phase of the CSEP (2007–2018) was mainly concentrating on the ‘homogeneous’ models of seismicity and deformation, with emphases on statistical seismology. And now, it is time to take more account of the tectonic settings and earthquake rupture processes.

For a certain segment of active fault or tectonic block boundary zone which would accommodate future earthquakes, an ‘earthquake rupture scenario’ plays the role of the ‘coordination’. This ‘earthquake rupture scenario’ provides different versions of earthquake preparation model. Based on the ‘earthquake rupture scenario’, detection programs and monitoring systems are to be designed, with consensus-based guidelines of experiments.

The guidelines should include several issues: (1) evaluating the ‘technical readiness level (TRL)’ of researches to be integrated into the experiment; (2) technical standard (and communication protocol) of data and data products to be shared (and cited); (3) planning for the possible earthquakes within (and surrounding) the test area/s; and (4) planning of the experiments, based on the activity (and uncertainty) of earthquakes, considering the ‘lifetime management’ of the observation/monitoring system, to ensure the effectiveness of the experiment.

In the planning of the experiment, several questions need to be carefully answered:

Q1: how many earthquakes we are expecting in the test area within the next few years?

⁴Here we use the words of late Prof. Leon Knopoff who commented world seismicity as ‘SO but not C’ in which SO stands for self-organized and C criticality.

Q2: what is the characteristic time scale of a geodynamical phenomenon for an observational system to capture?

Q3: how long is needed for an observation system to be deployed, calibrated, and 'pre-heated' for getting the baseline/background information?

Q4: what is the key evidence to be observed to test/falsify a model which predicts that something should be 'in this way'?

In one word, the effective constraint of the long-term seismogenic model and the short-term models for earthquake preparation, the effective observation/monitoring of 'expected' phenomena, and the effective test of the forecast schemes are the objectives of such coordination.

4.5 Coordination for a Specific Scientific Problem

One of the advantages of the CDEs is that, by experiments run in parallel by several research teams in multiple locations around the globe, scientific hypotheses can be addressed and tested, which requires the research teams to identify general and specific research questions.

For example, in the predictive modeling of earthquake cycles (e.g., Barbot et al. 2012), estimation of fault slip rates (as well as their uncertainty) based on geology and geodesy (e.g., Bennett 2007; Zechar and Frankel 2009) is an important issue. Data assimilation (Werner et al. 2011) in earthquake science is just at its beginning, but will play an important role in obtaining 'correct' estimates of the fault slip rates, which will facilitate the approaches to physical/numerical earthquake forecast. Such a scientific issue is in need of investigation making use of the observation/research resources in the experiment site/s.

Associated with the earthquakes occurred in the test sites, controversies in earthquake science could be resolved by testing against the real situation of earthquakes and seismic disasters. For example, whether the probabilistic quantification of time-dependent seismic hazard, such as that used in the Uniform California Earthquake Rupture Forecast (UCERF, Field et al. 2014, 2017), useful? And comparing to the probabilistic approach, whether the neo-deterministic seismic hazard assessment (NDSHA) approach (e.g., Zuccolo et al. 2010; Nekrasova et al. 2014) does a better job? The earthquakes will test. However, to make the test effective, the criterion and scheme for the test have to be specified beforehand.

4.6 Concluding Remarks and Discussion

The establishment of the CSES is underway. What is the chance to 'meet' a strong earthquake in this experiment site? The exact answer is 'we do not know'. However, according to the Chinese earthquake catalog, since 1965, there were on average 14

earthquakes with $M_S \geq 6.0$, among which 3 with $M_S \geq 7.0$, for every decade in this region, which provides the planning of the experiment with a statistical background.

Associated with this experiment site, there are several outstanding scientific problems to be addressed, making this experiment attractive. One example is that, in this continental region, are there any kind of low-frequency tremors which have been widely observed and studied in the plate boundary regions (Schwartz and Rokosky 2007; Ide et al. 2007; Beroza and Ide 2011; Kano et al. 2018), and (if exist) what is the role of such tremors in the seismogenesis process of continental earthquakes? Another example is that, comparing to other faults such as the Anatolian fault and the San Andreas fault (e.g., Kaneko et al. 2013; Tong et al. 2015), the creeping state of faults in this continental region (such as the Xianshuihe fault, the Anninghe fault, the Zemuhe fault, and the Xiaojiang fault) are still in lack of thorough investigation.

In the top-level design of the experiment sites, there are many issues to be considered. However, the most important challenge of such experiments is still ‘few samples, many factors’. Tackling this difficulty, we introduced the concept Coordinated Distributed Experiments (CDEs) from ecology and environmental science to earthquake science. We pointed out that these disciplines share some common languages in the design of the field experiments, although in ecology and environmental science the difficulty is mainly ‘spatial’, and in earthquake science the difficulty is mainly ‘temporal.’ In earthquake science, the experiment sites have long been ‘Distributed’, but one of the issues in need of more emphasizing is the ‘Coordination’. We proposed to use an ‘earthquake rupture scenario’ as the basis of the ‘coordination’. We suggested that there is a need for not only one but also several experiment sites. The China Seismic Experiment Site (CSES), as well as other existed (and being built) test sites, may play the role of pilot test site/s, with the missions of forming the guidelines for the Coordination.

Acknowledgements Thanks to Drs. Paramesh Banerjee and Li Li, President and Secretary-General of the Asian Seismological Commission (ASC), for invitation to the 12th ASC General Assembly in connection to the 4th ICCE, and to Prof. Yong-Gang Li for invitation to the current monograph.

References

- Audet, P. 2010. Temporal variations in crustal scattering structure near Parkfield, California, using receiver functions. *Bulletin of the Seismological Society of America* 100: 1356–1362.
- Bakun, W.H., and A.G. Lindh. 1985. The Parkfield, California, earthquake prediction experiment. *Science* 229: 619–624.
- Barbot, S., N. Lapusta, and J.P. Avouac. 2012. Under the hood of the earthquake machine, toward predictive modeling of the Seismic Cycle. *Science* 336: 707–710. <https://doi.org/10.1126/science.1218796>.
- Bennett, R.A. 2007. Instantaneous slip rates from geology and geodesy. *Geophysical Journal International* 169: 19–28.
- Beroza, G.C., and S. Ide. 2011. Slow earthquake and non-volcanic tremor. *Annual Review of Earth and Planetary Sciences* 39: 271–296. <https://doi.org/10.1146/annurev-earth-040809-152531>.

- Chao, K., and Z. Peng. 2009. Temporal changes of seismic velocity and anisotropy in the shallow crust induced by the 1999 October 22 M6.4 Chia-Yi, Taiwan earthquake. *Geophysical Journal International* 179: 1800–1816.
- Davis, J.F., and P. Somerville. 1982. Comparison of earthquake prediction approaches in the Tokai area of Japan and in California. *Bulletin of the Seismological Society of America* 72: S367–S392.
- Evans, R., D. Beamish, S. Crampin, and S.B. Ucer. 1987. The Turkish dilatancy project (TDP3), multidisciplinary studies of a potential earthquake source region. *Geophysical Journal of Royal Astronomical Society* 91: 265–286.
- Field, E.H., T.H. Jordan, M.T. Page, K.R. Milner, B.E. Shaw, T.E. Dawson, G.P. Biasi, T. Parsons, J.L. Hardebeck, A.J. Michael, R.J. Weldon II, P.M. Powers, K.M. Johnson, Y.H. Zeng, K.R. Felzer, N. van der Elst, C. Madden, R. Arrowsmith, M.J. Werner, and W.R. Thatcher. 2017. A synoptic view of the third Uniform California Earthquake Rupture Forecast (UCERF3). *Seismological Research Letters* 88: 1259–1267. <https://doi.org/10.1785/0220170045>.
- Field, E.H., R.J. Arrowsmith, G.P. Biasi, P. Bird, T.E. Dawson, K.R. Felzer, D.D. Jackson, K.M. Johnson, T.H. Jordan, C. Madden, A.J. Michael, K.R. Milner, M.T. Page, T. Parsons, P.M. Powers, B.E. Shaw, W.R. Thatcher, R.J. Weldon II, and Y.H. Zeng. 2014. Uniform California Earthquake Rupture Forecast, version 3 (UCERF3)—The time-independent model. *Bulletin of the Seismological Society of America* 104: 1122–1180. <https://doi.org/10.1785/0120130164>.
- Fraser, L.H., H.A.L. Henry, C.N. Carlyle, S.R. White, C. Beierkuhnlein, J.F. Cahill Jr., B.B. Casper, E. Cleland, S.L. Collins, J.S. Dukes, A.K. Knapp, E. Lind, R. Long, Y. Luo, P.B. Reich, M.D. Smith, M. Sternberg, and R. Turkington. 2012. Coordinated distributed experiments, an emerging tool for testing global hypotheses in ecology and environmental science. *Frontier in Ecology and Environment Science* 11: 147–155.
- Hoshiba, M. 2006. Current strategy for prediction of Tokai earthquake and its recent topics. http://cais.gsi.go.jp/UJNR/6th/orally/O04_UJNR_Hoshiba.pdf. Last access: 31 July 2018.
- Ide, S., G.C. Beroza, D.R. Shelly, and T. Uchide. 2007. A scaling law for slow earthquakes. *Nature* 447: 76–79.
- Kaneko, Y., Y. Fialko, D.T. Sandwell, X. Tong, and M. Furuya. 2013. Interseismic deformation and creep along the central section of the North Anatolian Fault (Turkey), InSAR observations and implications for rate-and-state friction properties. *Journal of Geophysical Research* 118: 316–331. <https://doi.org/10.1029/2012JB009661>.
- Kano, M., N. Aso, T. Matsuzawa, S. Ide, S. Annoura, R. Arai, S. Baba, M. Bostock, K. Chao, K. Heki, S. Itaba, Y. Ito, N. Kamaya, T. Maeda, J. Maury, M. Nakamura, T. Nishimura, K. Obana, K. Ohta, N. Poiata, B. Rousset, H. Sugioka, R. Takagi, T. Takahashi, A. Takeo, Y. Tu, N. Uchida, Y. Yamashita, and K. Obara. 2018. Development of a slow earthquake database. *Seismological Research Letters* 89: 1566–1575. <https://doi.org/10.1785/0220180021>.
- Lachenbruch, A.H., J.H. Sass, G.D. Clow, and R. Weldon. 1995. Heat flow at Cajon Pass, California. *Journal Geophysical Research* 100: 2005–2012.
- Mogi, K. 2004. Two grave issues concerning the expected Tokai Earthquake. *Earth, Plants and Space* 56: li–lxvi. <https://doi.org/10.1186/bf03353074>.
- Mora, P., and D. Place. 1998. Numerical simulation of earthquake faults with gouge: Toward a comprehensive explanation for the heat flow paradox. *Journal Geophysical Research* 103: 21067–21089. <https://doi.org/10.1029/98JB01490>.
- Nekrasova, A., V. Kossobokov, A. Peresan, and A. Magrin. 2014. The comparison of the NDSHA, PSHA seismic hazard maps and real seismicity for the Italian territory. *Natural Hazards* 70: 629–641. <https://doi.org/10.1007/s11069-013-0832-6>.
- Roeloffs, E. 2000. The Parkfield California earthquake experiment, an update in 2000. *Current Science* 79: 1226–1236.
- Roeloffs, E., and L. Langbein. 1994. The earthquake prediction experiment at Parkfield, California. *Review of Geophysics* 32: 315–336.
- Schorlemmer, D., M.J. Werner, W. Marzocchi, T.H. Jordan, Y. Ogata, D.D. Jackson, S. Mak, D.A. Rhoades, M.C. Hirata, N. Gerstenberger, M. Liukis, P.J. Maechling, A. Strader, M. Taroni, S. Wiemer, J.D. Zechar, and J. Zhuang. 2018. The Collaboratory for the Study of Earthquake Pre-

- dictability: achievements and priorities. *Seismological Research Letters* 89: 1305–1313. <https://doi.org/10.1785/0220180053>.
- Schwartz, S.Y., and J.M. Rokosky. 2007. Slow slip events and seismic tremor at circum-Pacific subduction zones. *Review of Geophysics* 45: RG3004. <https://doi.org/10.1029/2006RG000208>.
- Semenov, A.M. 1969. Variations in the travel-time of transverse and longitudinal waves before violent earthquakes. *Izvestia Academy of Science. USSR Physics of Solid Earth* (Engl. Transl.) 4: 245–248.
- Stefansson, R., F. Bergerat, M. Bonafede, R. Boovarsson, S. Crampin, K.L. Feigl, F. Roth, F. Sigmundsson, and R. Slunga. 2001. *PRENLAB-TWO—Final report*. Veourstofa Islands Report.
- Stefansson, R., F. Bergerat, M. Bonafede, R. Boovarsson, S. Crampin, P. Einarsson, K.L. Feigl, A. Guomundsson, F. Roth, F. Sigmundsson, and R. Slunga. 1998. *PRENLAB final report*. Reykjavik.
- Tong, X., D.T. Sandwell, and B.R. Smith-Konter. 2015. An integral method to estimate the moment accumulation rate on the Creeping Section of the San Andreas Fault. *Geophysical Journal International* 203: 48–62. <https://doi.org/10.1093/gji/ggv269>.
- Wegler, U., H. Nakahara, C. Sens-Schönfelder, M. Korn, and K. Shiomi. 2009. Sudden drop of seismic velocity after the 2004 Mw6.6 mid-Niigata earthquake, Japan, observed with passive image interferometry. *Journal of Geophysical Research: Solid Earth* 114: B06305. <https://doi.org/10.1029/2008jb005869>.
- Werner, M.J., K. Ide, and D. Sornette. 2011. Earthquake forecasting based on data assimilation: sequential Monte Carlo methods for renewal point process. *Nonlinear Processes in Geophysics* 18: 49–70. <https://doi.org/10.5194/npg-18-49-2011>.
- Zechar, J.D., and K.L. Frankel. 2009. Incorporating and reporting uncertainties in fault slip rates. *Journal Geophysical Research* 114: B12407. <https://doi.org/10.1029/2009JB006325>.
- Zuccolo, E., F. Vaccari, A. Peresan, and G.F. Panza. 2010. Neo-deterministic and probabilistic seismic hazard assessments: A comparison over the Italian territory. *Pure and Applied Geophysics* 168: 69–83. <https://doi.org/10.1007/s00024-010-0151-8>.